

ASSIGNMENT 4.4

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating Process – Step by Step

1. Power Off and Unplug



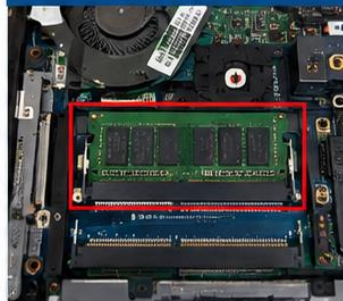
I shut down the laptop, unplugged the charger and removed the battery (if possible) for safety.

2. Open the Laptop Back Panel



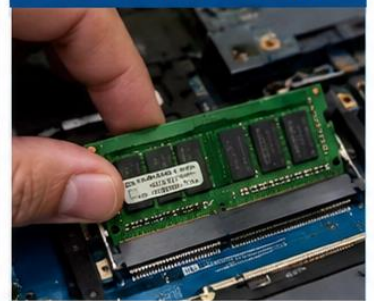
I removed the screws from the back panel and carefully took off the cover to access the internal components.

3. Locate the RAM Slot



I found the RAM slot (it was easy to identify — the module is a small green stick with clips on both sides).

4. Remove the RAM Module



I pushed the side clips outward gently. The RAM module popped up at an angle, so I pulled it out carefully.

5. Check the RAM Module



I checked the RAM module for dust, damage or bent pins. It looked fine.

6. Reinsert the RAM Module



I aligned the notch on the module with the slot and pressed it down until the clips locked it in place.

7. Close the Laptop



I put the back panel back on and tightened the screws.

8. Power On and Test



I plugged in the charger and turned on the laptop. It booted normally and everything seemed to work fine.

✓ Issues Observed

- No major issues were observed during the process.
- The RAM module came out and went back in smoothly.
- The system booted normally after reseating.

💡 Final Result

RAM reseating was successful. The laptop is working properly and is running without any errors.

Before you start

- Turn off the PC and unplug the power cable.
- Press and hold the power button for 10 seconds.
- Use a screwdriver and work on a clean table.
- Touch a metal part of the case to remove static.

Steps to reseat RAM

1. Open the PC case or laptop back cover.
2. Find the RAM sticks next to the CPU fan.
3. Push the clips on both sides of the RAM to release it.
4. Take the RAM out by holding the edges (do not touch the gold pins).
5. Wipe the gold pins gently with a dry, soft cloth.
6. Match the notch on the RAM with the slot and push it in firmly.
7. Listen for a click. The clips should lock by themselves.
8. Close the case, plug in power, and turn on the PC.

Common issues and what to do

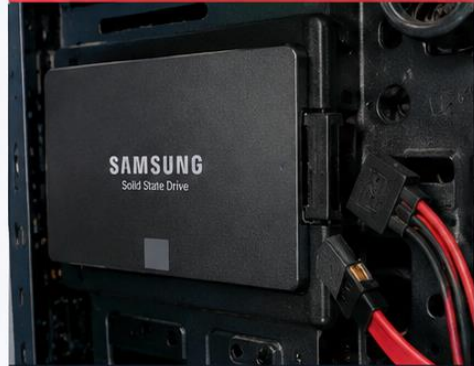
- PC does not start: check if RAM is fully pushed in and in the right direction.
- Beep sound or memory error on screen: try one RAM stick at a time.
- BIOS shows less RAM: reseat the RAM again and check in BIOS.
- Always ground yourself before touching RAM.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.
Hint: Think about display issues, driver checks, and physical connections.

- 1. Check symptoms:** no display, strange colors/lines, game crashes, black screen, or driver errors.
- 2. Check cables and power:** make sure GPU is tight in the slot and all power cables are connected.
- 3. Test monitor and cable:** try another cable or monitor. If CPU has video output, try that too.
- 4. Update or reinstall GPU driver:** remove old driver, then install the latest driver from the company website.
- 5. Run a GPU test:** use a stress test app and watch temperatures. If problems continue, GPU may be faulty.

3.Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

1. Unplug the SATA Drive (simulate issue)



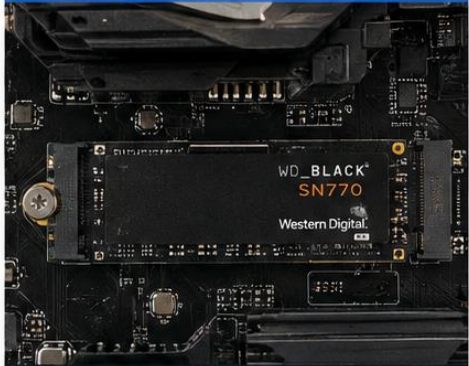
SATA data cable disconnected from SSD.

2. Replug the SATA Drive



SATA data cable reconnected to SSD.

(Optional) NVMe Drive Example



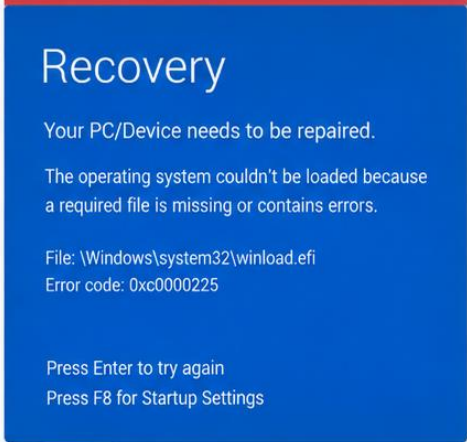
NVMe SSD (M.2) shown for reference.

BIOS (Before Reconnecting)



BIOS shows the SATA device as "Not Detected" (after unplugging the drive).

Before Reconnecting)



Windows shows a boot error (when the drive is disconnected).

BIOS (After Reconnecting)



BIOS now detects the SSD/HDD correctly (after reconnecting the drive).

Issues Observed (Before Reconnecting)

- **BIOS:** Drive shown as "Not Detected" or missing from boot list.
- **Windows:** Boot error (e.g., 0xc0000225) or "No boot device" message.
- **System:** May fail to boot or show a black screen.

Final Result (After Reconnecting)

- **BIOS:** Drive appears in the storage/boot list.
- **Windows:** Boots normally and detects the drive.
- **System:** Drive is accessible and works as expected.

When drive is unplugged

- **BIOS:** drive not shown in SATA or NVMe list.
- **If it was the main drive:** “No boot device” or “Operating system not found”.
- **Windows:** drive letter missing in File Explorer.
- **Disk Management:** disk missing or shown as “Unknown”. When drive is plugged back in
- **BIOS:** drive appears again in SATA/NVMe list.
- **Windows:** drive letter shows up again (for data drives).
- **If cable is loose:** drive may come and go, or show “Disk read error”.
- **If NVMe is not tight:** BIOS may still not see it. Reseat and tighten the screw.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

First component to check

First check: RAM

Why RAM first?

Bad RAM can corrupt data and make the PC restart suddenly. RAM tests are easy and fast.

Steps to check RAM

- 1. Run Windows Memory Diagnostic:** search “Windows Memory Diagnostic”, run it, and restart. Check results in Event Viewer.
- 2. Run MemTest86:** make a bootable USB, boot from it, and run at least 4 passes. Any error means bad RAM.
- 3. Test one stick at a time:** if you have two or more sticks, test each one alone.
- 4. Check BIOS:** if you use XMP/overclock, turn it off and use default RAM speed.
- 5. Reseat RAM:** power off, take out RAM, and put it back in firmly. Then test again. If

RAM is OK

Next check GPU (drivers, artifacts, stress test), storage (SMART health, errors), power supply, and temperatures.